

HEALTHCONNECT CLINIC

Analytics Testing & Refinement Report

Data Analytics Track

Introduction

Week 7 focused on testing, refinement and end-to-end validation of the HealthConnect Analytics work developed in Weeks 5 and 6. The work concentrated on independently checking important calculations, reviewing dashboard behaviour and interpretation, refining the presentation where appropriate, and documenting the evidence.

Week 6 to Week 7 Transition

Week 6 extended the HealthConnect analysis from exploratory reporting into deeper relationship analysis and decision support. The outputs carried into Week 7 included the KPI set, Booking Lead analysis, Previous No-Show analysis, Reminder Channel analysis, and combined analyses using Booking Lead, Previous No-Show History and Reminder Channel.

Week 7 therefore focused on validating these existing outputs against the underlying HealthConnect appointment data and reviewing whether the dashboard communicated the findings clearly and consistently.

Testing Approach

The underlying HealthConnect appointment data in Excel was used as the independent validation source. Excel calculations were compared with the corresponding Power BI results. Testing covered KPI calculations, segmented no-show calculations, selected matrix values, dashboard filters, and visual usability. Where the expected result was achieved, the test was recorded as a successful validation.

KPI Validation

KPI Tested	Expected Result	Actual Result	Result
No-Show Rate	Excel calculation should agree with Power BI	Values agreed	PASS
Attendance Rate	Excel calculation should agree with Power BI	Values agreed	PASS
Cancellation Rate	Excel calculation should agree with Power BI	Values agreed	PASS
Reminder-Associated Attendance Rate	Excel calculation should agree with Power BI	Values agreed	PASS

Repeat No-Show Rate	Excel calculation should agree with Power BI	Values agreed	PASS
---------------------	--	---------------	------

The Week 6 dashboard displayed an overall No-Show Rate of approximately 48.5%, while the supporting total in the Week 6 analysis was 48.46%. The difference is consistent with rounding.

Analytical Validation

Component	Testing Activity	Result
Appointment Type	Checked General Consultation, Follow-up, Diagnosis and Specialist Consultation calculations.	PASS
Booking Lead Group	Checked 0–7, 8–14, 15–30 and 31+ day groups.	PASS
Previous No-Show Group	Checked previous no-show categories/groups against the underlying records.	PASS
Reminder Channel	Checked channel-level attendance/no-show calculations.	PASS
Waiting Time Group	Checked waiting-time segments, including the 60+ minute category.	PASS
Selected matrices	Checked selected cells in Booking Lead × Previous No-Show and Previous No-Show × Reminder Channel.	PASS

Validated Findings

- Booking Lead Group showed different observed no-show rates across the analysed groups, with the highest displayed rate in the 31+ day group.
- Previous No-Show History generally showed higher observed no-show rates as prior no-show history increased; extreme percentages were interpreted with appointment volume in mind.
- The Booking Lead × Previous No-Show analysis provided deeper segmentation than either variable alone.
- Attendance/No-show rates differed across reminder channels in the analysed dataset.

Testing & Validation Report

This record shows the principal tests performed. PASS means the expected result was achieved. Where no issue was found, no artificial issue or correction is recorded.

Testing Matrix

Component Tested	Expected	Actual	Result	Issue	Action
No-Show Rate KPI	Excel and Power BI should agree	Agreed	PASS	None	No correction required
Attendance Rate KPI	Excel and Power BI should agree	Agreed	PASS	None	No correction required
Cancellation Rate KPI	Excel and Power BI should agree	Agreed	PASS	None	No correction required
Reminder-Associated Attendance Rate KPI	Excel and Power BI should agree	Agreed	PASS	None	No correction required
Repeat No-Show Rate KPI	Excel and Power BI should agree	Agreed	PASS	None	No correction required
Appointment Type	Each category should agree with Excel	Agreed	PASS	None	No correction required
Booking Lead Group	Each group should agree with Excel	Agreed	PASS	None	No correction required
Previous No-Show Group	Each group should agree with Excel	Agreed	PASS	None	No correction required
Reminder Channel	Channel results should agree with Excel	Agreed	PASS	None	No correction required
Waiting Time Group	Segment results should agree with Excel	Agreed	PASS	None	No correction required
Booking Lead × Previous No-Show	Selected cells should agree with Excel	Agreed	PASS	None	No correction required
Previous No-Show × Reminder Channel	Selected cells should agree with Excel	Agreed	PASS	None	No correction required
Dashboard Filters	Filters should update relevant visuals correctly	Responded as expected	PASS	None	No correction required
Dashboard Usability	Visuals should be clear, non-redundant and decision-support oriented	Reviewed and refined where appropriate	PASS	Presentation refinement	Refined selected visuals/filter hierarchy

Dashboard Testing and Refinement

The dashboard was reviewed for calculation consistency, duplication, readability, filter behaviour and decision-support value. Refinement focused on improving hierarchy and reducing repetition rather than rebuilding the Week 6 work.

Component	Testing/Refinement Decision
Core KPI cards	Retained and validated.
Booking Lead analysis	Retained as a major decision-support visual.
Previous No-Show analysis	Retained and interpreted with appointment volume context.
Booking Lead × Previous No-Show	Retained as a key deeper analysis component.
Previous No-Show × Reminder Channel	Retained as a supporting interaction analysis.
Slicers	Reviewed so that filters retained clear analytical purpose.

Retesting

After validation and dashboard review, important outputs were checked again to confirm that the analytical values remained consistent and that the refined presentation still communicated the intended findings. Successful validations were recorded as successful.

Business Insights

- Booking lead time is associated with different observed no-show rates across the analysed groups.
- Previous no-show history is associated with higher observed no-show rates.
- Combining booking lead and previous no-show history provides more detailed segmentation for appointment risk analysis.
- Observed attendance rates differ across reminder channels; these results should be interpreted as associations rather than proof of causation.
- Long waiting-time appointments, particularly the 60+ minute category, warrant monitoring with appropriate volume context.

Recommendations

- Use booking lead time and previous no-show history as practical segmentation variables when monitoring appointment risk.
- Consider appointment volume alongside percentages, especially for small or extreme subgroups.
- Continue monitoring reminder channel outcomes without treating observed differences as causal effects.
- Monitor long waiting-time appointments and investigate the operational factors behind unusually high observed no-show rates.
- Maintain a focused dashboard hierarchy that prioritises core KPIs and the strongest analytical relationships.

Cross-Track Testing Contribution

The validated Analytics outputs provide structured inputs for wider HealthConnect solution development. Variables include booking lead time, previous no-show history, reminder channel and their interaction.

I documented the validated findings and the proposed variables for future Data Science testing. This provides a structured handoff for assessing feature relevance, interaction effects, and potential contribution to a predictive no-show model in a subsequent phase.

Analytics contribution:

- Validated key HealthConnect analytical findings.
- Identified potential modelling variables from the validated analysis.
- Highlighted the Booking Lead Time $\times$ Previous No-Show History relationship for further investigation.
- Documented the proposed testing areas for future Data Science integration.

Limitations, Risks and Dependencies

- Observed relationships do not establish causation.
- Small subgroups can produce extreme percentages, so volume should be considered alongside rates.
- The analysis is limited to variables available in the HealthConnect dataset.
- Reminder-channel comparisons may reflect differences in the appointments receiving each channel.

Conclusion

Week 7 demonstrated progression from development to evidence-based validation. The main analytical calculations were checked against the underlying data, important relationships were validated, dashboard usability and redundancy were reviewed, and the decision support presentation was refined while preserving the established HealthConnect work.